SPMM Smart Revise

Neuroanatomy

Paper A **Syllabic content 3.1**

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1. General anatomy of the brain

A. Cortical structures

The cerebrum has four major lobes (frontal, temporal, parietal and occipital lobes). The lobar surface is heavily folded forming sulci (valleys) and gyri (ridges). **Primary (major) sulci** are more invariant in their appearance than the **secondary (minor) sulci**.

The **central sulcus** divides frontal lobe from the parietal lobe. Precentral gyrus (part of the frontal lobe) is the **primary motor cortex**. The representation of different body parts in this region is often termed as a homunculus. Postcentral gyrus (part of the parietal lobe) is the **primary somatosensory cortex** with a similar homunculus representation.

The **lateral sulcus** (Sylvian fissure) divides frontal lobe from the temporal lobe. The **insula**, a structure that is sometimes regarded as the fifth lobe of the cerebrum, is located deep in the Sylvian fissure. Insula is the seat of the primary gustatory cortex.

Other major primary sulci include

1. Superior and inferior frontal sulci: In between these sulci is the middle frontal gyrus constituting the **dorsolateral prefrontal cortex**, often considered to be responsible for executive functions of the human brain.
2. Cingulate sulcus on the medial side of the frontal lobe. The anterior portion of the adjoining **cingulate gyrus** is considered to be the seat of motivation.
3. Olfactory and orbital sulci on the inferior surface of the frontal lobe. The **orbitofrontal cortex** is often considered to be the seat of associative learning and decision-making.
4. The Superior temporal sulcus is forming superior temporal gyrus, the seat of **primary auditory cortex**.
5. The interparietal sulcus separates superior and inferior parietal lobes. **The inferior parietal lobe** is made of the angular gyrus and supramarginal gyrus and is considered to be important for visuospatial attention.
6. Calcarine sulcus in the medial occipital cortex, the seat of **primary visual (striate) cortex**

Hemispheric lateralisation

- Most fundamental brain functions are represented bilaterally. Higher levels of associative functions usually lateralize to one or other hemisphere. For example, language comprehension is localized to the left temporal cortex while prosody (tonal modulation of speech) seems limited to the right hemisphere.
- The hemisphere contralateral to the dominant hand is the dominant hemisphere, and it mediates language and speech functions.
- Dominance can be tested using Annette's handedness scale or Edinburgh handedness inventory. But **handedness** is not always same as dominance.

- In right-handed people, the left hemisphere is mostly dominant. In 10% of right-handed people, the right hemisphere is dominant. Among left-handed people only about 20% are right hemisphere dominant, with 64% left hemisphere dominant and 16% showing bilateral dominance.
- Size asymmetry: The **planum temporale** is a triangular region on the upper surface of the superior temporal gyrus. It is important for language processing and is larger on the left than the right hemisphere in 65% brains. It is probably the most asymmetrical structure in the human brain, with some individuals having a five times larger planum temporale on the left than on the right. This asymmetry is reportedly reduced or reversed (right>left) in schizophrenia.

Left Hemisphere lesions	Right Hemisphere lesions
Aphasia	**Visuospatial deficits**
Right-left disorientation	**Anosognosia**
Finger agnosia	**Neglect**
Dysgraphia (aphasic)	**Dysgraphia (spatial, neglect)**
Dyscalculia (number alexia)	**Dyscalculia (spatial)**
Limb apraxia	**Constructional apraxia** **Dressing apraxia**
Face recognition (bilateral)	

B. Subcortical structures

Limbic system/ Papez circuit

- Broca first described the limbic lobe. Papez and later Maclean assigned the function of emotional processing to limbic structures though this view is challenged in recent times.
- The Papez circuit consists of the hippocampus → fornix → mammillary bodies → mammillothalamic tract → anterior thalamic nucleus → genu of the internal capsule → cingulate gyrus → parahippocampal gyrus → entorhinal cortex → perforant pathway → back to hippocampus
- The boundaries of the limbic system were subsequently expanded outside of the Papez circuit to include the amygdala, septum, basal forebrain, nucleus accumbens, and orbitofrontal cortex.
- The limbic system is thought to be involved in various functions such as mediation of emotional responses (through amygdala), influencing neuroendocrine responses (via hypothalamus) and reward system regulation (via nucleus accumbens).
- The limbic system is often considered to be evolutionarily older than the higher cortical centres.

Medial temporal structures

- Include the hippocampus, amygdala, entorhinal and parahippocampal cortex.
- Hippocampus appears to play an important role in memory processes. It is one of the few brain regions where the continuous production of new neurons is noted even in adult life.
- Amygdala appears crucial for fear conditioning and emotional regulation

Basal Ganglia

- The basal ganglia are a group of gray matter nuclei forming the largest subcortical structure in the brain. They are involved in the planning and programming of movement, and also have a role in the processes by which an abstract thought is converted into voluntary action

- They include **striatum** made of the ***caudate nucleus*** and ***putamen*** and **pallidum** made of ***globus pallidus*.** Putamen and globus pallidus are sometimes called ***lenticular/lentiform nucleus***.
- The ***subthalamic nuclei*** and the ***substantia nigra*** are both functionally related to the basal ganglia but are not considered to be a part of this structure.
- Basal ganglia receive crucial inputs from glutamatergic ***corticostriatal*** projection. Alexander described five important circuits involving the basal ganglia. These are
    - Motor circuit
    - Oculomotor circuit
    - Dorsolateral prefrontal circuit (executive)
    - Anterior cingulate circuit (motivation)
    - Lateral orbitofrontal circuit (social intelligence)

Disorder	Nature of basal ganglia dysfunction
OCD	**Volumetric changes and higher blood flow to the caudate nuclei. Increased caudate metabolism in untreated subjects reduces after effective treatment.**
Tourette's syndrome	**Striatal dopaminergic dysfunction**
Huntington chorea	**Degeneration of the striatum (mainly caudate nucleus) & selective loss of GABAergic neurons**
Wilson disease	**Copper deposits in the lenticular nuclei**
CO poisoning	**Acute bilateral anoxic damage to basal ganglia**
Hemiballismus	**Subthalamic nucleus damage (especially infarction)**
Parkinsonism	**Depigmentation of Substantia Nigra; Lewy bodies are seen. Striatal overactivity associated with bradykinesia**
Fahr's disease	**Progressive calcium deposition in the basal ganglia. (early onset cases present with schizophreniform psychoses and catatonia; later onset cases exhibit dementia and choreoathetosis)**

Thalamus

- A large oval mass of grey matter nuclei in the subcortical region, relaying all types of sensory information onto cortex (except olfaction).
- It also relays cerebellar and basal ganglia inputs to the cerebral cortex.
- The thalamus is said to play a crucial role of filtering sensory information in preparation for cortical processing.
- The anterior thalamus is a part of the limbic system. It receives the mamillothalamic tract and fornix and connects to the cingulate cortex. Thus, it relays information from hypothalamus and hippocampus onto the frontal cortex.
- Pulvinar is associated with visual attention. Sleep spindles are generated in the reticular nucleus of the thalamus.

Hypothalamus

- The hypothalamus regulates physiological functions such as eating, drinking, sleeping, and temperature regulation.
- The hypothalamus has chemoreceptors that respond to variations in glucose levels, osmolarity, acid balance, etc. It also plays a major role in neuroendocrine control.
- *The ventromedial hypothalamus* acts as the satiety centre while the *lateral hypothalamus* is the feeding centre. In animals with a lesion of ventromedial hypothalamus hyperphagia and obesity are noted.

INFERIOR OLIVARY NUCLEUS

Inferior olivary nucleus *is located in the brainstem and aids in motor coordination by projecting climbing fibers to the contralateral cerebellar cortex via inferior cerebellar peduncle.*

Inferior olivary lesions lead to appendicular ataxia due to motor incoordination of the contralateral arm and leg. Patients with inferior olivary lesions will fail the finger-nose test, mimicking cerebellar lesion. But unlike cerebellar lesions that result in ipsilateral motor incoordination, the contralateral side is affected in olivary lesions.

C. Cerebellum

The cerebellum has the important role of preparing a motor plan and predicting balance needed between muscle groups to carry out the intended action smoothly. Cerebellar lesions produce ataxia and coarse intentional tremors, along with hypotonia, past pointing and pendular knee jerk.

Increasingly the role of the cerebellum in cognitive processes has been appreciated. The term **cognitive dysmetria** (Andreasen) refers to the difficulty in coordinating and monitoring the process of receiving, processing, and expressing information that could result from disrupted cortico-cerebellar circuitry in schizophrenia.

D. Brain stem and cranial nerves

The brain stem is made of the midbrain, pons and the medulla. Most of the cranial nerves (9 out of 12) enter or exit the brain from the brainstem.

The midbrain consists of **superior** (conjugate gaze control) and **inferior colliculi** (auditory source localization). **The substantia nigra** is also located in the midbrain along with **periaqueductal grey matter** that plays an important role in vocalization and freezing response to threat and in pain suppression.
Pons is positioned beneath the cerebellum and surrounds the upper half of the 4th ventricle
Medulla surrounds inferior part of the 4th ventricle and is continuous with the spinal cord.

No.	Name	Anatomical features
I	Olfactory	Runs on the basal surface of frontal cortex without passing through the thalamus. Formed as an outgrowth of forebrain
II	Optic	Also an outgrowth of the forebrain. Relays via thalamus (geniculate body)
III	Oculomotor	Purely motor function. Supplies four of the six ocular muscles
IV	Trochlear	Purely motor function. Supplies superior oblique (ocular muscle)
V	Trigeminal	Both sensory and motor. Transmits facial sensation and controls jaw muscles
VI	Abducens	Purely motor function. Supplies lateral abducens (ocular muscle)

VII	Facial	Both sensory and motor. Transmits taste sensation and controls facial muscles
VIII	Vestibular	Transmits auditory sensation
	Cochlear	Transmits balance sensation
IX	Glossopharyngeal	Motor control of pharynx; parasympathetic control of the parotid gland; taste from the back of the tongue.
X	Vagus	Motor control of larynx and pharynx; parasympathetic control of the viscera; visceral sensations.
XI	Accessory	Motor control of neck muscles
XII	Hypoglossal	Motor control of tongue muscles

E. Spinal Cord

Unlike cerebrum where grey matter is on the outer surface, in spinal cord grey matter occupies the deeper aspect forming an H shaped column surrounding the CSF. The white matter bundles form anterior, lateral and dorsal columns around the grey matter zone. The **dorsal column** carries proprioceptive sensory fibres; the anterior and lateral columns are made of ascending **spinothalamic tracts** carrying touch, pressure, pain and temperature sensations.

F. Cerebrospinal fluid

CSF is secreted by the choroid plexus in the lateral, third and fourth ventricles and at a rate of 300 ml/day, which is almost protein free.

Route: From lateral ventricle to 3rd ventricle via interventricular **foramina of Monroe**; From 3rd to 4th ventricle via cerebral **aqueduct of Sylvius**; From 4th ventricle to subarachnoid space via **foramen of Magendie** (single) and **foramina of Luschka** (two lateral).

The body of the lateral ventricle lies immediately below the corpus callosum, and the two lateral ventricles are separated by septum pellucidum. The third ventricle lies between thalamus and hypothalamus. The fourth ventricle lies above the pons and just below the cerebellum.

Obstruction to CSF circulation commonly occurs within third or fourth ventricle (foramen of Monroe), leading to **non-communicating hydrocephalus**. Impairment of CSF reabsorption in the subarachnoid space due to partial occlusion of the arachnoid villi leads to **communicating hydrocephalus.**

2. Blood supply to the brain

A. Major branches

The internal carotid artery enters the circle of Willis and divides to form the anterior cerebral and middle cerebral arteries.

The **anterior cerebral artery** supplies the medial and superior strip of the lateral aspect of the cerebral cortex up to the parietal/occipital border.

The **middle cerebral artery** supplies most of the lateral aspect of the cerebral cortex. This includes the Broca's and Wernicke's areas in the dominant hemispheres.

The **posterior cerebral artery** arises from basilar artery and supplies the inferomedial temporal lobe and the occipital lobe.

The medulla is supplied by **posterior inferior cerebellar arteries** and anterior spinal branches of vertebral arteries.

Pons is supplied by the **basilar artery** that runs along the midline of the pons.

Carotid system TIA	Vertebrobasilar TIA
• Amaurosis fugax (due to blockade of retinal arteries) • Aphasia • Hemiparesis • Hemisensory loss • Hemianopic visual loss	• Diplopia, vertigo, vomiting • Choking and dysarthria • Ataxia • Alexia without agraphia • Hemisensory loss • Hemianopic visual loss • Transient global amnesia • Tetraparesis • Loss of consciousness (rare)

B. Effect of lesions

Artery	Supply	Lesion effects
Anterior Cerebral Artery (ACA)	Medial surface (ventromedial frontal lobe, the cingulum, the premotor cortex, and medial motor strip)	Bilateral infarct produces quadriparesis (legs weaker than arms) and akinetic mutism (ventromedial or cingulate syndrome)
Recurrent artery of Huebner (branch of ACA)	Head of the caudate nucleus	Initially an agitated, confused state; evolves to akinesia, abulia, with mutism and personality changes
Anterior branches of the upper division of the Middle Cerebral Artery	Lateral prefrontal cortex	Planning deficits, impairment of working memory, and apathy. (DLPFC dysfunction)
Anterior communicating artery	Basal forebrain	Akinesia and personality change (orbitofrontal dysfunction) with a confabulatory amnesia resembling Wernicke-Korsakoff syndrome.
Posterior inferior cerebellar artery (PICA) thrombosis	Lateral medulla	Wallenberg's lateral medullary syndrome. Acute vertigo with cerebellar signs. Ipsilateral face numbness, diplopia, nystagmus, Horner's syndrome and IX/X nerve palsy with contralateral spinothalamic sensory loss and mild hemiparesis.

3. White matter pathways

There are 3 major types of white matter pathways. **Projection fibers** run vertically connecting higher and lower centres of the brain. **Association fibers** interconnect different regions within the same hemisphere of the brain. **Commissural fibers** interconnect similar regions in the opposite hemisphere.

Corpus callosum is the largest bundle of fibres that connect the two cerebral hemispheres; the other such bundles are **anterior commissure** (interconnects olfactory bulbs)**, posterior commissure** (interconnects midbrain pretectal nuclei)**, hippocampal commissure** and **habenular commissure** (interconnects posterior dorsal thalamic nuclei).

The pericallosal artery derived from the anterior cerebral artery provides blood supply to the anterior aspect and most of the body of the corpus callosum**. Left sided apraxia and agnosia** may be seen in cases of vascular disruption.

Posterior cerebral artery territory supplies **splenium** (posterior aspect of the corpus callosum) and disrupted supply here prevents right visual cortex accessing the dominant hemispheric processes such as language resulting in **alexia and color anomia** but with preserved ability to copy words as motor information is relayed via anterior corpus callosum

Fornix is an important white mater tract that connects hippocampus to the hypothalamus via mammillary bodies. Thus, it relays cortical input to regulate neuroendocrine and autonomic systems.

Arcuate fasciculus connects Broca's and Wernicke's areas. Damage results in conduction aphasia.

Uncinate fasciculus is a major frontotemporal tract that connects orbitofrontal cortex to the anterior temporal lobes. It plays an important role in social cognition and language.

4. Cell types in the nervous system

A. Cortical layers

The human brain contains approximately 10^{11} neurons (nerve cells) and approximately 10^{12} glial cells.

According to the distribution of the various types of neurons (i.e. the cytoarchitecture), Brodmann divided the cortex into 47 'specialised' areas.

The neocortex (most of the cerebrum) is made up of six layers, with pial surface above layer 1 to the white matter below layer 6. Layers 2 and 4 are mainly afferent (receiving inputs) while 5 and 6 are mainly efferent (sending outputs).

The pyramidal neurons with their triangular-shaped cell bodies make up nearly 75% of the cortical neurons. Stellate cells (25%) are present in all the layers except layer 1.

Layer	Name	Predominant cells
1	**Molecular/agranular**	**Glial cells, dendrites from neurons of deeper layers and the horizontal cells of Cajal.**
2	**External Granular layer**	**Granule cells and small pyramidal cells (these get larger as you move down)**
3	**External Pyramidal layer**	**Small and medium sized pyramidal cells.**
4	**Internal Granular layer**	**Some pyramidal cells, mostly granule cells. Receives thalamocortical inputs.**
5	**Internal Pyramidal layer**	**Largest pyramidal cells (esp. in motor cortex: Betz cells)**
6	**Multiform layer**	A mixture of all cells, **spindle cells, Martinotti cells.** The major **source of corticothalamic fibres. Gives rise to association/commissural and projection fibres.**

The **cerebellar cortex** is three layered. The **molecular** layer consisting of basket cells and stellate cells, **Purkinje** layer consisting of Purkinje cells and a **Granular** layer consisting of granule and Golgi cells.

B. Special neuronal cell types

Purkinje cells are a class of GABAergic neurons located in the cerebellar cortex only. Purkinje cells form the sole output of all motor coordination in the cerebellum they connect to the deep cerebellar nuclei via inhibitory projections.

Granule cells are found within the granular layer of the cerebellum, layer 4 of the cerebral cortex, the dentate gyrus of the hippocampus, and in the olfactory bulb.

Large pyramidal cells called **Betz cells** are seen in the primary motor cortex. Betz cells are pyramidal cell neurons located within the fifth layer of the grey matter in the primary motor cortex. These neurons are

the largest in the central nervous system, sometimes reaching 100 μm in diameter. Betz cells represent about 10% of the total pyramidal cell population in layer V of the human primary motor cortex.

Stellate cells are found in layer IV of the cerebral cortex (from thalamus feeding forward to pyramidal cells) and also in the cerebellum.

C. Glial cells

These are cells with supportive metabolic functions; they also participate in modulating neuronal functions e.g. via the production of neurosteroids. There are 3 types of glial cells:

1. ***Astrocytes*** are the most numerous of the three types. These are star-shaped cells that enable nutrition of neurons, breakdown of some neurotransmitters, and maintaining the blood-brain barrier.
2. ***Oligodendrocytes*** are seen in CNS (not in peripheral nerves, where Schwann cells replace them). They produce myelin sheaths that help in saltatory conduction (pole to pole jumping), which quicken the process of signal transmission.
3. The ***microglia*** are descendants of macrophages. They are scavenger cells that clear neuronal debris following cell death.
4. ***Ependymal cells*** are a special type of glia that cover the ventricles and facilitate CSF circulation via their ciliary processes.

BLOOD BRAIN BARRIER

*The blood- brain barrier is located in **endothelial cells** of capillaries of the brain. Unlike the endothelial cells found elsewhere, brain's endothelial cells have **tight junctions** with high electrical resistance providing an effective barrier against molecules. In addition, brain capillaries are in contact with foot processes of **astrocytes** that separate the capillaries from the neurons.*

Lipid soluble molecules (ethanol and caffeine) can penetrate the barrier relatively easily via the lipid membranes of the cells. In contrast, water-soluble molecules such as sodium and potassium ions are unable to transverse the barrier without using specialized carrier-mediated transport mechanisms.

Inflammation such as meningitis weakens the blood brain barrier.

*There are some areas of the brain that do not have a blood- brain barrier. These are so called **circum-ventricular organs** e.g. subfornical organ, area postrema (chemo receptor trigger zone), median eminence and posterior pituitary.*

5. Major neurochemical pathways

A. Dopaminergic pathways

Depending on the length of the projections, dopaminergic pathways can be classified into

1. *Long paths*: **Nigrostriatal**, **mesocortical** and **mesolimbic** pathways.
2. *Short paths*: **Tuberoinfundibular** and **incertohypothalamic** pathway.
3. *Ultrashort paths*: These are found in the amacrine cells in the retina and in the olfactory system.

The nigrostriatal pathway is the extrapyramidal pathway that is crucial for motor control; this accounts for most of the brain's dopamine.

Pathway	Origin and destination	Effect of dopamine (DA) blockade
Nigrostriatal	Substantia Nigra to striatum and amygdala via medial forebrain bundle	**DA deficiency (e.g Parkinson's) or blockade due to antipsychotics can cause extrapyramidal side effects**
Mesolimbic	Ventral tegmental area (VTA) to Nucleus accumbens and hippocampus via medial forebrain bundle	**Blockade of DA in this tract produces the desirable antipsychotic effect by controlling positive psychotic symptoms**
Mesocortical	Ventral tegmental area (VTA) to cingulate cortex and prefrontal regions via medial forebrain bundle	**Low levels of DA or DA blockade in this tract is associated with negative symptoms (alogia, anhedonia, amotivation and apathy)**
Tuberoinfundibular	Hypothalamus to the pituitary via portal vessels	**Dopamine acts as PIH – prolactin inhibitory hormone. DA blockade will serve to increase prolactin levels.**
Incertohypothalamic	**Internal connections within hypothalamus**	**Disturbed thermoregulation and possibly weight gain**

B. Cholinergic pathways

The two major cholinergic pathways are

1. *Brainstem pathway*: This forms a part of the ARAS - ascending reticular activating system that is important to maintain wakefulness and REM sleep state. It originates from **pedunculopontine** and laterodorsal tegmental nuclei and innervates thalamic relay neurons and reticular nuclei.
2. *Basal forebrain pathway*: Originates at the **Nucleus Basalis of Meynert** in basal forebrain and projects to the hippocampus, frontal cortex and amygdala. Degeneration of this pathway is implicated in **Alzheimer's disease.**

C. Serotonergic pathways

Most of the brain's serotonergic neurons originate in the midbrain **dorsal and median raphe** nuclei and ascend to innervate the entire cortex, basal ganglia, thalamus, and also descend to the spinal cord.

D. Noradrenergic pathways

Noradrenergic projection originates at the locus coeruleus (pons) and ascends to most of the cortex via medial forebrain bundle. Similar to the serotonin system, noradrenergic projections also descend to the spinal cord.

E. Glutamatergic system

Glutamate is the most common excitatory neurotransmitter in the brain. As a result, almost all cortical descending tracts (from pyramidal cells) rely on glutamate for neurotransmission. This large output of **corticofugal fibres** makes up most of the **corona radiata.** All of the association and commissural fibres also use glutamatergic transmission. Many thalamic neurons are also glutamatergic. Thus **thalamocortical** projections are also glutamatergic. In addition cerebellar output from deep nuclei, subthalamic nuclei to globus pallidus projections, and brainstem to spinal cord projections are also predominantly glutamatergic.

F. GABAergic system

GABA is the primary inhibitory neurotransmitter in the brain. Unlike other neurotransmitters, there are no specific neurochemical pathways where GABA is dominant. Instead, GABA is the major transmitter for cerebral interneurons that are ubiquitous throughout the cortex.

Interneurons are usually short neurons that serve to connect two other neurons, thus forming an essential part of the complex wiring pattern of the cortex. They carry neither motor nor sensory information but serve to modulate local neural circuitry.

2 major cortical interneuron subtypes are noted: **parvalbumin (PV)-expressing interneurons** (~40% of all interneurons) and **somatostatin (SST)-expressing interneurons** (30% of interneurons, also called Martinotti cells).

A reduction in the expression of PV-interneurons in the frontal cortex is now a well-replicated feature of schizophrenia. PV-interneurons are of 2 subtypes: Basket cells and Chandelier cells.

Basket cells receive direct input from thalamocortical projections. They form synapses with the soma or dendrites of the pyramidal neurons and serve to provide the excitatory-inhibitory balance to the cortex.

Chandelier cells form synapses with the proximal axonal hillock of pyramidal neurons. They may have an overall excitatory role by serving to short-circuit the action potential propagation though their role is still unclear.

DISCLAIMER: This material is developed from various revision notes assembled while preparing for MRCPsych exams. The content is periodically updated with excerpts from various published sources including peer-reviewed journals, websites, patient information leaflets and books. These sources are cited and acknowledged wherever possible; due to the structure of this material, acknowledgements have not been possible for every passage/fact that is common knowledge in psychiatry. We do not check the accuracy of drug related information using external sources; no part of these notes should be used as prescribing information.

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